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A novel cyathane diterpene, cyrneine A, induces neurite outgrowth in a Rac1-dependent mechanism in PC12 cells

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Abstract

We examined the effects of cyrneine A, a novel diterpene isolated from the mushroom *Sarcodon cyrneus*, on morphology of rat pheochromocytoma cells (PC12). In response to cyrneine A, PC12 cells extended their neurites, an effect partially blocked by the extracellular signal-regulated kinase (ERK) kinase inhibitor PD98059, but not by the protein kinase C inhibitor GF109203X, nor the phosphatidylinositol-3-kinase inhibitor wortmannin. Cyrneine A did not activate ERK at any of the time points tested (5–120 min), indicating that only the basal activity of ERK is required for cyrneine A-induced neurite outgrowth. As transcriptional regulation is required for neurite extension, the activity of three major transcription factors was determined. Cyrneine A enhanced activation of the transcription factors activator protein-1 (AP-1) and nuclear factor-κB, but not CREB, and this was accompanied by enhanced *c-fos* expression. Moreover, we determined the role of Rac1, a small GTPase protein of the Rho family that regulates actin dynamics, in cyrneine A-induced neurite outgrowth. Treatment with cyrneine A led to actin translocation and subsequently, to accumulation of F-actin at the tip of neurites. Rac1 activity was increased by cyrneine A and expression of a dominant-negative Rac1 mutant significantly inhibited the cyrneine A-induced extension of neurites. These results suggest that cyrneine A induces neurite outgrowth in a Rac1-dependent mechanism.
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Keywords: Cyrneine; *Sarcodon cyrneus*; Rac; Neurite outgrowth; PC12 cells

Introduction

Neurotrophic factors, including nerve growth factor (NGF), have a pleiotropic effect on the promotion of neuronal differentiation and survival in various neurons (Levi-Montalcini, 1987). In light of their beneficial effects, neurotrophic factors were suggested as potential therapeutic agents for the treatment of neurodegenerative conditions such as Alzheimer's or Parkinson's disease. However, neurotrophic factors are polypeptides that cannot cross the blood–brain barrier; a useful strategy for addressing this drug delivery problem is the use of small organic compounds which might directly improve neuronal function or up-regulate neurotrophic factors. To date, small molecules such as AIT-082 and SR57746A have been shown to have NGF-like agonistic properties or to induce the production and secretion of NGF (Appert-Collin et al., 2004; Grundman et al., 2003).

Cyathane diterpenes represent an interesting group of complex molecules, the first of which was isolated in 1971 from the mushroom *Cyathus helenae* (Allbutt et al., 1971). Several other cyathane diterpenes have since been isolated from *Cyathus* sp. (Hecht et al., 1978) and mushroom varieties such as *Hericium erinaceum* (Kawagishi et al., 1994), *Sarcodon scabrosus* (Ohta et al., 1998) and *Sarcodon glaucopus* (Curini et al., 2005). Many of these compounds have interesting biological properties, such as antibacterial (Anke and Oberwinkler, 1977) and anti-inflammatory activities (Hirota et al.,

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2002; Kamo et al., 2004). Furthermore, cyathanes have attracted attention for their stimulatory action on the biosynthesis of neurotrophic factors in vitro (Kawagishi et al., 1994). Previously, we showed that scabronines isolated from *S. scabrosus*, promoted the secretion of neurotrophic factors, including NGF, from 1321N1 human astrocytoma cells, causing rat pheochromocytoma cells (PC12 cells) to differentiate (Obara et al., 1999, 2001). The total synthesis of a scabronine derivative, scabronine G-methylester, was recently achieved and was shown to have the same pharmacological effects as naturally derived scabronines (Waters et al., 2005; Obara et al., 1999, 2001).

We have screened a number of compounds isolated from medicinal plants in order to develop NGF-like drugs. Previously, we found that the novel *Sarcodon cyrneus* cyathane diterpenes, cyrneines A and B, induce neurite outgrowth in the PC12 cell model of neuronal differentiation (Marcotullio et al., 2006). Here, we report that cyrneine A enhances neurite outgrowth in a Rac1-dependent mechanism in PC12 cells.

Materials and methods

Materials

Cyrneine A was isolated from *S. cyrneus* as described (Marcotullio et al., 2006). NGF, dibutyryl cAMP (dbcAMP), wortmannin, UTP and the RNA extraction kit were purchased from Sigma Aldrich (St. Louis, MO). Paraformaldehyde, phorbol-12-myristate-13-acetate (PMA), GF109203X, anisomycin and forskolin were purchased from Wako Pure chemicals (Tokyo, Japan). Antibodies against protein kinase C (PKC) isoforms (α, δ and ζ) were from Santa Cruz (Santa Cruz, CA). CGS21680 was from Tocris Cookson Ltd. (Bristol, UK). Antibodies against phospho-extracellular signal-regulated kinases (ERKs), total ERKs, phospho-c-Jun-N-terminus kinase (JNK), phospho-p38 mitogen-activated protein kinase (MAPK), phospho-ERK5, phospho-cAMP-response element-binding protein (CREB) and horseradish peroxidase (HRP)-conjugated anti-rabbit IgG antibody, PD98059 and U0126 were purchased from Cell Signaling (Beverly, MA). Anti-Rac1 was purchased from BD Biosciences, Japan (Tokyo, Japan). [^{3}H] inositol (23.4 Ci/mmol) was from DuPont/NEM (Boston, MA). The Anion exchange column (AG1X8) and Dowex 50W-X8 were from BioRad (Hercules, CA). The Alumina column was from Merck (Darmstadt, Germany). The DNA plasmids (Elk-1/Gal 4 and 5 × Gal 4-E1B-luciferase) were a kind gift from Dr. Philip Stork (Vollum Institute, Oregon Health Sciences University, Portland, OR). Reporter plasmids for measuring transcriptional activity (pTRE/luciferase, pCRE/luciferase, pNF-κB/luciferase) and pCMV-enhanced green fluorescein protein (EGFP) were from Clontech (Palo Alto, CA). DNA plasmids encoding a dominant-negative Rac1 mutant (Rac1T17N or RacN17) and a glutathione-S-transferase-tagged p21-activated kinase 1-Cdc42/Rac interactive binding region (GST-PAK1-CRIB) were a kind gift from Dr. Thomas Soderling (Vollum Institute, Oregon Health Sciences University). Glutathione-sepharose, HRP-conjugated anti-mouse IgG antibody, the enhanced chemiluminescence assay kit and [^{3}H]adenine were all from Amersham Pharmacia (Buckinghamshire, England). The reverse transcription-polymerase chain reaction (RT-PCR) kit was obtained from Toyobo (Osaka, Japan) and lipofectamine 2000 was purchased from Invitrogen (Grand Island, NY). Phalloidin-rhodamine was from Molecular Probes (Eugene, OR) and the luciferase assay kit was from Promega (Madison, WI).

Cell culture

PC12 cells were obtained from the Japanese Cancer Research Bank (Tokyo, Japan). PC12 cells were grown in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% heat-inactivated fetal calf serum (Cell Culture Laboratory, Cleveland, OH), 5% horse serum (Invitrogen), penicillin (50 units/ml), and streptomycin (50 μg/ml) in an incubator maintained at 37 °C with 5% CO_2. NG108-15 cells were cultured in DMEM supplemented with 10% fetal calf serum, hypoxanthine (100 μM), aminopterin (1 μM), thymidine (16 μM) and the above antibiotics.

Evaluation of neurite outgrowth

PC12 cells were incubated with cyrneine A for 24–48 h in DMEM supplemented with 0.5% horse serum and 1% fetal calf serum. PC12 cells were fixed with 4% paraformaldehyde in phosphate-buffered saline (PBS), and cell morphology was observed under a differentiation-interference microscope. Neurite extension in PC12 cells is regarded as an index of neuronal differentiation; in this study, processes with a length equivalent to one or more diameters of the cell body were regarded as neurites. The differentiation of PC12 cells was evaluated by examining the proportion of neurite-bearing cells to the total number of cells in a randomly selected area. The mean differentiation score was obtained for more than one hundred PC12 cells in each of three wells that the cells were seeded onto, and data are expressed as mean ± S.E.M (Obara et al., 2002). The neurite outgrowth assay for NG108-15 cells was carried out in DMEM supplemented with 1% fetal calf serum, hypoxanthine, aminopterin and thymidine.

Subcellular fractionation

Samples for PKC detection were prepared as follows: PC12 cells were seeded onto a 6-well plate at a density of 8×10^5 cells/well. The medium was replaced with serum-free DMEM and the cells were cultivated overnight. After incubation with cyrneine A, homogenization buffer (50 mM Tris–HCl, 5 mM EDTA, 5 mM EGTA, 0.3% β-mercaptoethanol, 1 mM phenylmethylsulfonyl fluoride, 10 μg/ml leupeptin, 10 μg/ml aprotinin, 1 μg/ml antipain, pH 7.5) was added to terminate the reaction. After sonication (5 s × 5 times), cells were homogenized and centrifuged at 100,000 ×*g* for 60 min at 4 °C. The resulting supernatant was considered as the cytosolic fraction. The pellets were dissolved in homogenization buffer containing 1% triton X-100 and incubated for 30 min at 4 °C, before centrifuging at 100,000 ×*g* for 30 min at 4 °C. The resulting supernatant was regarded as the total membrane fraction and the

remaining pellets, as the cytoskeletal fraction (Trilivas et al., 1991). Fractions were dissolved in Laemmli sample buffer (final concentration, 75 mM Tris–HCl, 2% SDS, 15% glycerol, 3% 2-mercaptoethanol, pH 6.8) and boiled at 95 °C for 5 min.

SDS-polyacrylamide gel electrophoresis and Western blotting

Samples were prepared as follows: PC12 cells were seeded onto 24-well plates at a density of 2×10^5 cells/well. The medium was replaced with serum-free DMEM and the cells were cultivated overnight. Cells were incubated with cyrneine A in serum-free EMEM–HEPES for a variety of time-scales. The incubation medium was aspirated after the reaction and cells were dissolved in Laemmli sample buffer, and boiled at 95 °C for 5 min. Electrophoresis was performed on 11% acrylamide gels. Proteins were transferred from the gel onto polyvinylidene difluoride membrane (Millipore, Bedford, MA) by semi-dry blotting. Membranes were blocked for 1 h with 2% low-fat milk in Tris-buffered saline containing 0.1% tween-20 (TBST) at room temperature, and incubated with primary antibodies overnight at 4 °C. Membranes were washed several times in TBST before incubating with an HRP-conjugated anti-rabbit or anti-mouse IgG diluted in TBST containing 2% low-fat milk, at room temperature for 2 h. After rinsing with TBST, membranes were developed using a chemiluminescence assay kit, and bands were visualized by exposing the membrane to Hyper-film ECL.

Transfection and reporter gene assay

The reporter plasmids (Elk–gal, TRE, CRE or NF-κB plasmids) were transfected into PC12 cells using Lipofectamine 2000. Briefly, the cells were seeded onto 48-well plates at 1×10^5 (cells/well) and cultivated for a day. The plasmids (0.5 μg/tube) and transfection reagent (1 μl/tube) were mixed gently in DMEM (10 μl/tube) and incubated for 20 min at room temperature. After the addition of DMEM (40 μl/tube), the mixture (50 μl/well) was added to the culture media, which had been replaced with serum-free DMEM (200 μl). The cells were incubated for 5 h at 37 °C before the serum-free media was replaced with growth medium containing 10% fetal calf serum and 5% horse serum again. After serum starvation, cells were then incubated with drugs at 37 °C for 6 h and a luciferase assay was performed according to the manufacturer's instruction. Luciferase activity was measured using a luminometer (GENE LIGHT 55, Microtech Nition, Funabashi, Japan).

Measurement of inositol phosphates

The accumulation of total inositol phosphates was measured as follows: PC12 cells were cultivated in 12-well plates for two days at a density of 2×10^5 cells/ml/well, then incubated in medium containing [^{3}H] inositol (2 μCi/ml) overnight. After washing twice with serum-free EMEM–HEPES buffer (pH 7.4), the cells were preincubated in EMEM–HEPES containing LiCl (10 mM) for 10 min, before incubating with drugs for additional 10 min. The reaction was terminated by aspirating the medium and replacing it with 1 ml of ice-cold 5% trichloroacetic acid. Trichloroacetic acid extracts were washed three times with diethylether to remove all traces of trichloroacetic acid and then incubated at 47 °C for 30 min to remove all traces of diethylether. Total [^{3}H] inositol phosphates were separated in an anion exchange column (AG-1X8, formate form, 100–200 mesh) and radioactivity was quantified by liquid scintillation counting as previously described (Nakahata et al., 1989).

cAMP assay

Intracellular cAMP levels were determined as follows: PC12 cells were cultivated in 24-well plates and 2 μCi/ml of [^{3}H] adenine (23 Ci/mmol) was added to the growth medium for 3 h. The labeled cells were washed twice with DMEM-HEPES (20 mM, pH 7.4) and then incubated for 10 min with cyrneine A, forskolin or CGS21680 in DMEM-HEPES containing the phosphodiesterase inhibitor, Ro20-1724 (100 μM). After aspirating the incubation buffer, reactions were quenched by adding 0.4 ml of 2.5% $HClO_4$. Acid extracts were mixed with a 1/10 volume of 4.2 M KOH to neutralize the acid, forming potassium perchlorate as a precipitate. [^{3}H]cAMP in the supernatant was separated by Dowex 50 W-X8/alumina double columns, and radioactivity was quantified by liquid scintillation counting. The increase in [^{3}H]cAMP levels were expressed as fold change of control (no drug).

Semi-quantitative RT-PCR

Total RNA from PC12 cells was extracted using a total RNA extraction kit, and semi-quantitative RT-PCR was carried out using an RT-PCR kit. The sense (5′-GTT CTC GGG TTT CAA CGC GGA CTA CGA GGC-3′) and antisense primers (5′-GGC ACT AGA GAC GGA CAG ATC TGC GCA AAA GTC C-3′) were designed to a conserved region of rat *c-fos* cDNA. The resulting product (169 bp) was amplified by 28 cycles of: 95 °C for 60 s, 55 °C for 60 s and 72 °C for 60 s. Sense (5′-ATG ACT GCA AAG ATG GAA ACG-3′) and antisense primers (5′-TGC CGC GGA GGT GAC ACT GGG-3′), were designed to rat *c-jun* cDNA and the resulting product (402 bp) was amplified by 33 cycles of: 94 °C for 60 s, 56 °C for 60 s, 72 °C for 100 s. Glyceraldehyde 3-phosphate dehydrogenase (G3PDH) transcripts (700 bp) were used as an internal control. The optimum number of cycles (to yield a quantifiable amount of product) was determined in a previous experiment.

Affinity assay for Rac1 activation

A GST fusion protein of PAK 1's Rac1-binding domain (GST-PAK-CRIB) was expressed in *Escherichia coli* following induction by isopropyl-1-thio-β-D-galactopyranoside, and purified using glutathione-sepharose beads. Cells were stimulated at 37 °C for the indicated times in 35 mm dishes and lysed in ice-cold buffer (50 mM Tris–HCl (pH 7.4), 10% glycerol, 1% nonidet P-40, 200 mM NaCl, 2.0 mM $MgCl_2$, 500 μM β-glycerophosphate, 1 mM phenylmethylsulfonyl fluoride, 1 μg/ml leupeptin, 1 μg/ml antipain and 2 μg/ml aprotinin). A volume of cell lysate containing 500 μg of total protein was incubated

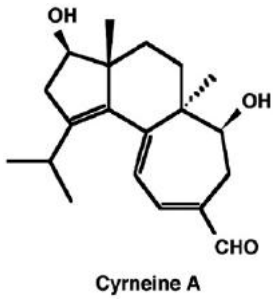

Fig. 1. Chemical structure of cyrneine A.

with GST-PAK-CRIB, coupled to glutathione beads (10 μl). Following a 30 min incubation at 4 °C, beads were pelleted and rinsed threes times with ice-cold lysis buffer. Proteins were eluted from the beads using Laemmli buffer before loading onto 15% SDS-polyacrylamide gels. Active Rac1 was detected by Western blotting using anti-Rac1 antibody.

Statistics

Data were expressed as mean ± S.E.M, and significant differences were analyzed using Dunnett's or Tukey's methods.

Results

The chemical structure of cyrneine A is shown in Fig. 1. PC12 cells extended their neurites in a concentration-dependent manner when incubated in cyrneine A (100 or 200 μM) for 24 h (Fig. 2A). As expected, neurite extension was also observed following treatment with NGF (100 ng/ml; Fig. 2A), however, we wanted to establish whether a combined treatment with cyrneine A and NGF had an additive or synergistic effect on neurite extension. Following incubation in NGF (100 ng/ml) and cyrneine A (200 μM) for 48 h, neurite outgrowth was observed in a greater number of cells, when compared to single treatment (Fig. 2B). To determine whether the effect of cyrneine A on neurite extension is PC12-cell-specific, we examined neurite outgrowth in NG108-15 cells, a hybrid neuronal cell line derived from mouse neuroblastoma and rat glioma. NG108-15 cells responded to a 24 h incubation in cyrneine A (100 or 200 μM) by significantly extending their neurites (Fig. 2C). NG108-15 cells have been shown to extend neurites in response to an increase in intracellular cAMP levels (Tojima et al., 2003). We therefore examined neurite extension following treatment with the membrane-permeable cAMP analog, dbcAMP, and observed a similar occurrence of neurite outgrowth (Fig. 2C).

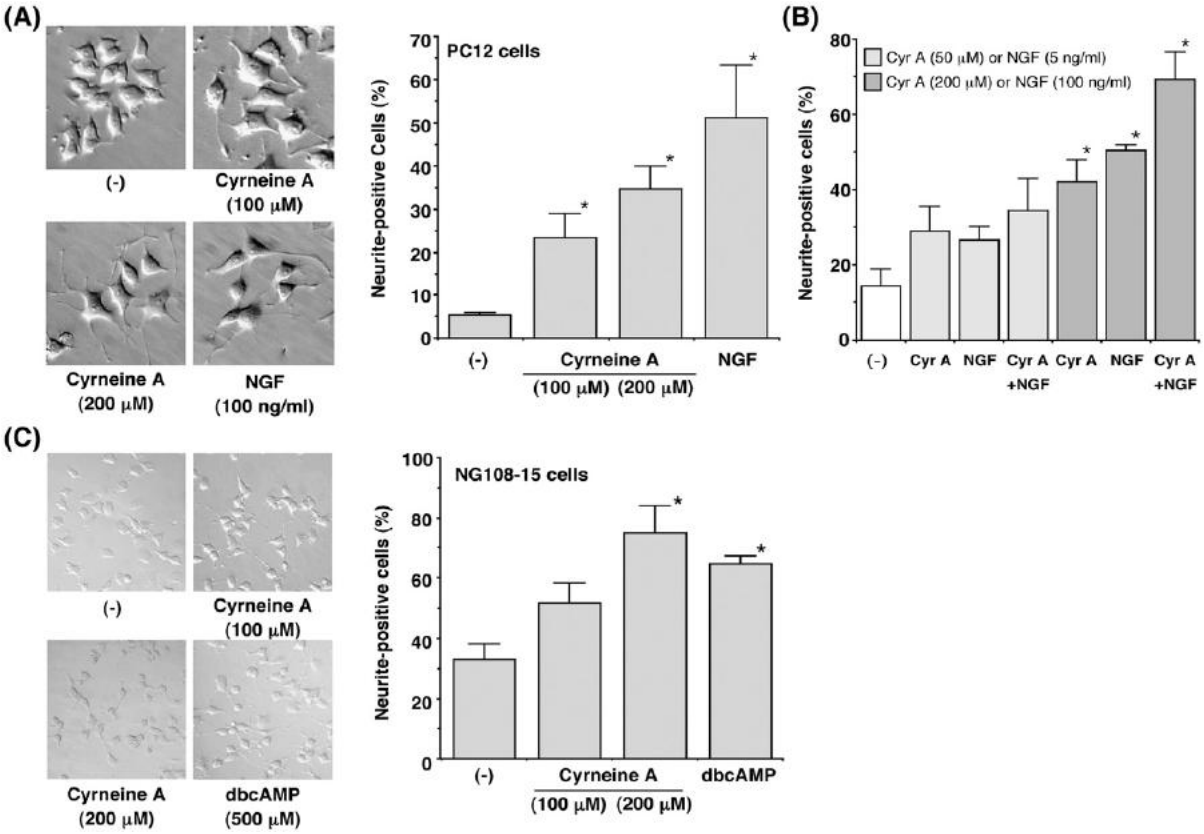

Fig. 2. Cyrneine A induces neurite outgrowth in PC12 cells and NG108-15 cells. (A) PC12 cells were incubated with cyrneine A (100 and 200 μM) or NGF (100 ng/ml) for 24 h, and morphological changes were observed. Cyrneine A (100 and 200 μM) and NGF (100 ng/ml) significantly induced neurite outgrowth (*$P<0.05$). (B) PC12 cells were incubated with cyrneine A (50 μM), NGF (5 ng/ml), cyrneine A (50 μM) + NGF (5 ng/ml), cyrneine A (200 μM), NGF (100 ng/ml) or cyrneine A (200 μM) + NGF (100 ng/ml) for 48 h. Cyrneine A (200 μM), NGF (100 ng/ml) and cyrneine A (200 μM) + NGF (100 ng/ml) significantly induced neurite outgrowth (*$P<0.05$). (C) NG108-15 cells were incubated with cyrneine A (100 and 200 μM) or dbcAMP (500 μM) for 48 h. Cyrneine A (200 μM) and dbcAMP (500 μM) significantly induced neurite outgrowth (*$P<0.05$). In all cases, the percentage of neurite-positive cells were determined from three cell populations and data are expressed as mean ± S.E.M.

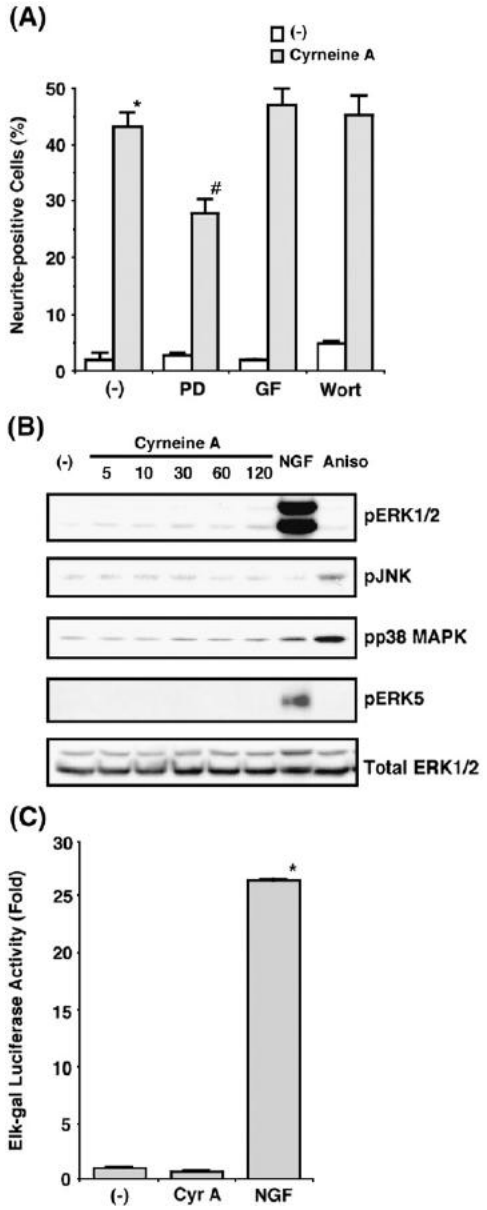

Fig. 3. Basal activity of ERKs is required for cyrneine A-induced neurite outgrowth in PC12 cells. (A) PC12 cells were preincubated with PD98059 (PD, 30 μM), GF109203X (GF, 10 μM) or wortmannin (Wort, 100 nM) for 30 min, before incubation with cyrneine A (200 μM) for 24 h. Cyrneine A significantly induced neurite outgrowth (*$P<0.05$), and this effect was significantly inhibited by preincubation with PD98059 ($^{\#}P<0.05$). The percentage of neurite-positive cells were determined from three cell populations and data are expressed as mean±S.E.M. (B) PC12 cells were incubated with cyrneine A (200 μM) for 5–120 min, NGF (100 ng/ml) for 5 min or anisomycin (Aniso, 30 μM) for 30 min. Phosphorylation of ERK 1/2, JNK, p38 MAPK, or ERK 5 and total ERK1/2 were examined by Western blotting. (C) PC12 cells were transfected with plasmids encoding Elk-1/Gal 4 and 5×Gal 4-E1B-luciferase and subsequently treated with cyrneine A (200 μM) or NGF (100 ng/ml) for 6 h. Elk-1 activation was monitored by luciferase activity. Data are expressed as mean±S.E.M. for $n=3$ cell populations. NGF, but not cyrneine A, significantly increased the in vivo activity of ERKs (*$P<0.05$).

To examine the mechanism of neurite outgrowth induced by cyrneine A (200 μM), we targeted a number of kinases using specific inhibitors (Fig. 3A). The MEK inhibitor PD98059 (30 μM) led to a significant but partial decrease in the number of neurite-positive cells, and a similar level of inhibition was observed with the MEK inhibitor U0126 (data not shown). However, the PKC inhibitor GF109203X (10 μM) and the phosphatidylinositol-3-kinase (PI3K) inhibitor wortmannin (100 nM) were without effect on neurite outgrowth following cyrneine A treatment (Fig. 3A). MEK operates upstream of ERK in the ERK pathway and since MEK inhibition led to a

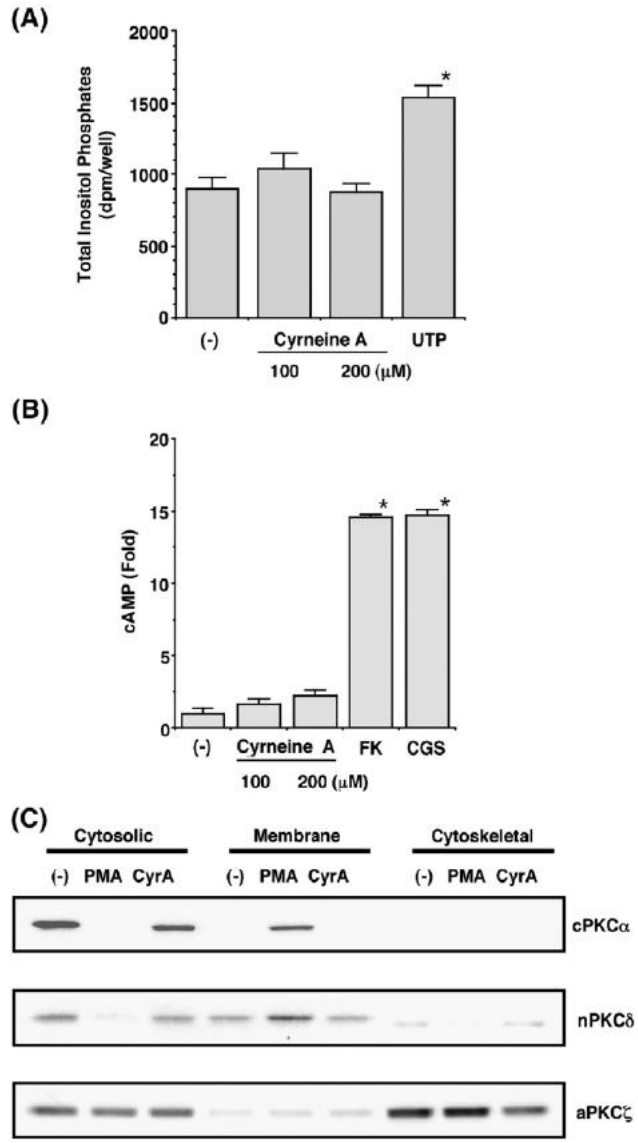

Fig. 4. Cyrneine A does not affect cyclic AMP or total inositol phosphates levels, nor does it cause translocation of PKC isoforms in PC12 cells. (A) PC12 cells were stimulated with cyrneine A (100 or 200 μM) or UTP (100 μM) for 10 min before the levels of total inositol phosphates were determined. Data are expressed as mean±S.E.M. for $n=3$ cell populations. UTP significantly promoted the accumulation of inositol phosphates (*$P<0.05$). (B) PC12 cells were stimulated with cyrneine A (100 or 200 μM), forskolin (FK, 10 μM) or CGS21680 (CGS, 10 μM) for 10 min, before the levels of cAMP were determined. Data are expressed as mean±S.E.M. for $n=3$ cell populations. Forskolin and CGS21680 significantly increased cAMP levels (*$P<0.05$). (C) PC12 cells were incubated with cyrneine A (Cyr A) (200 μM) or PMA (100 nM) for 30 min, and the cell homogenates were separated into cytosolic, membrane and cytoskeletal fractions. The content of conventional PKCα, novel PKCδ and atypical PKCζ in each fraction was investigated by Western blotting.

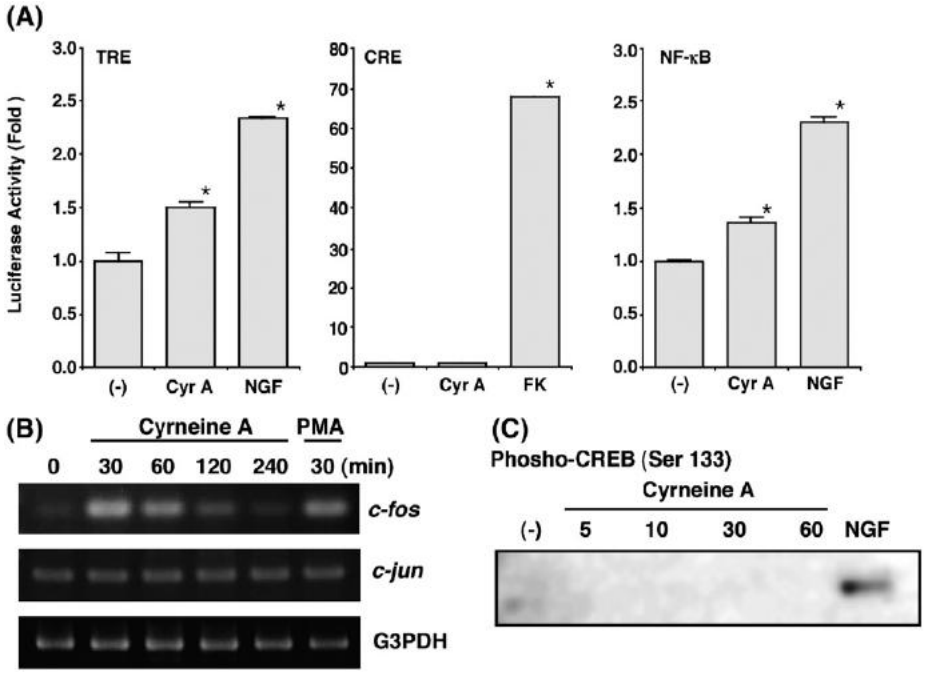

Fig. 5. Cyrneine A increases the activity of transcription factors AP-1 and NF-κB in PC12 cells. (A) PC12 cells were transfected with pTRE/luciferase, pCRE/luciferase or pNF-κB/luciferase reporter plasmids, and subsequently stimulated with cyrneine A (200 μM), NGF (100 ng/ml) or forskolin (10 μM) for 6 h. Transcriptional activity was monitored by measuring luciferase activity. Data are expressed as mean ± S.E.M. for $n = 3$ cell populations. Cyrneine A and NGF significantly increased activity of AP-1 and NF-κB (*$P < 0.05$) and forskolin significantly activated CREB (*$P < 0.05$). (B) PC12 cells were incubated with cyrneine A (200 μM) for a range of times or with PMA (100 nM) for 30 min and gene expression of *c-fos*, *c-jun* and G3PDH was examined by RT-PCR. (C) PC12 cells were stimulated with cyrneine A (200 μM) for a range of times or with NGF (100 ng/ml) for 5 min and phosphorylation of CREB at Ser 133 was observed by Western blotting.

reduction in the number of neurite-positive cells, we investigated whether treatment with cyrneine A led to activation of ERK in PC12 cells. Cells were stimulated with cyrneine A and the phosphorylation of MAPK family members was monitored at various time points as an index of activation (Fig. 3B). However, there was no change in the phosphorylation levels of ERK 1/2, JNK, p38 MAPK or ERK 5 following treatment with cyrneine A (200 μM) at any of the observed time points. Furthermore, we measured ERK activity in vivo with an Elk/Gal reporter system in cells incubated in cyrneine A (200 μM) for 6 h. Cyrneine A treatment did not lead to activation of ERK, whereas NGF (100 ng/ml) significantly increased ERK activity (Fig. 3C). Taken together, these data suggest that the mechanism of neurite outgrowth induced by cyrneine A does not occur through activation of the ERK pathway, however, the inhibition of outgrowth by the MEK inhibitor PD98059 suggests that basal ERK MAPK pathway activity is required.

In order to further characterise the mechanism of cyrneine A induced neurite outgrowth, we examined the phosphoinositide and cAMP second messenger systems. Phosphoinositide hydrolysis was significantly accelerated by the purinergic receptor agonist, UTP (100 μM), however, cyrneine A treatment (100 or 200 μM) was without effect (Fig. 4A). As expected, treatment with the adenylyl cyclase activator forskolin (10 μM) or the adenosine agonist CGS21680 (10 μM) led to a large increase in intracellular cAMP levels and although cyrneine A (100 or 200 μM) marginally increased cAMP levels, this was not significant (Fig. 4B). Furthermore, we examined PKC translocation following treatment with cyrneine A, as an index of PKC activation. We observed that the conventional PKC isoform α, and the novel PKC isoform δ, both translocated from

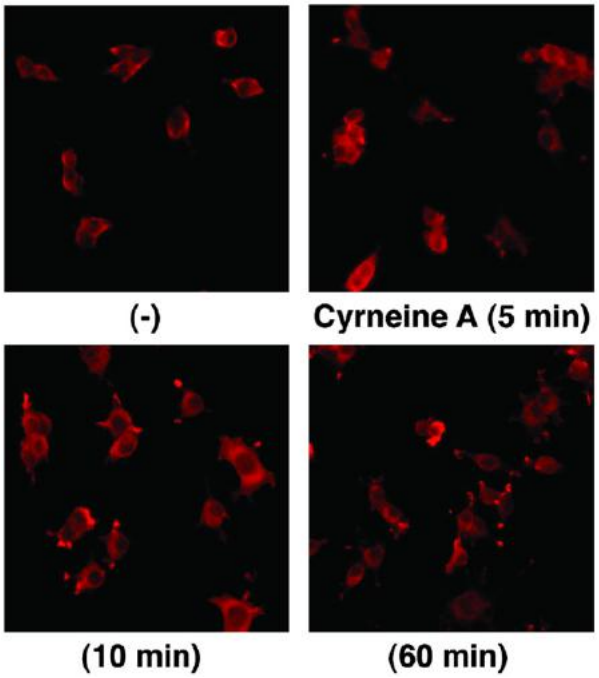

Fig. 6. Cyrneine A promotes rearrangement of actin in PC12 cells. PC12 cells were stimulated with cyrneine A (150 μM) for a range of times. Following fixation in 4% paraformaldehyde, F-actin filaments were stained with phalloidin-rhodamine and observed by fluorescence microscopy.

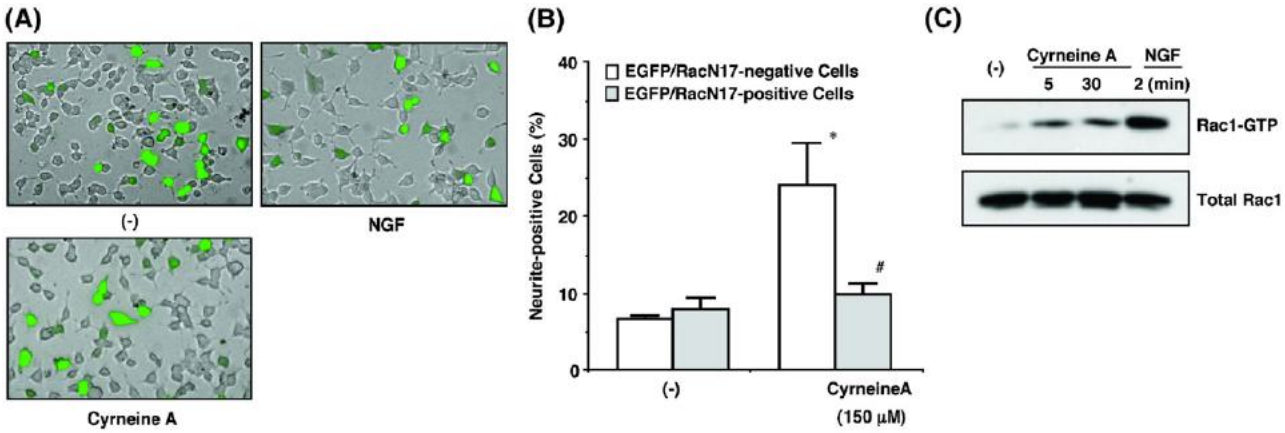

Fig. 7. Cyrneine A-induced neurite outgrowth is accompanied by activation of Rac1 in PC12 cells. (A) PC12 cells were co-transfected with EGFP and Rac1N17. Cells were incubated with cyrneine A (150 μM) or NGF (100 ng/ml) for 24 h, and morphological changes were observed under a fluorescence microscope. Green cells indicate EGFP and Rac1N17-expressing cells. (B) The neurite outgrowth assay was performed as (A), and the morphology of EGFP-positive and -negative cells were determined. Data are expressed as mean ± S.E.M. for $n=3$ cell populations. Cyrneine A significantly promoted neurite outgrowth ($^{*}P<0.05$), and this effect was significantly suppressed by Rac1N17 expression ($^{\#}P<0.05$). (C) PC12 cells were stimulated with cyrneine A (150 μM) for 5 and 30 min or with NGF (100 ng/ml) for 2 min, and subjected to pull-down with GST-PAK-CRIB, followed by Western blotting. Active Rac1 (Rac1-GTP) is shown in upper panel and total Rac1 is shown in lower panel. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the cytosolic fraction to membrane fraction in response to a 30 min treatment with PMA (100 nM; Fig. 4C). However, there was no overtly obvious translocation of any of the PKC isoforms tested in response to cyrneine A (200 μM), although a small amount of PKCζ appeared to translocate to the membrane fraction. These data, together with the apparent lack of effect on neurite outgrowth by the PKC inhibitor GF109203X, suggest that PKC is not involved in the mechanism of cyrneine A-induced neurite extension (Fig. 3A).

In response to growth factors such as NGF, cytoskeletal components, regulators (peripherin, cofilin and MAP 4) and signaling molecules (Trio, RhoG and p35), are up-regulated in response to activation of several transcription factors, and these are all assumed to be essential for neurite outgrowth (Angelastro et al., 2000; Estrach et al., 2002; Harada et al., 2001; Katoh et al., 2000; Troy et al., 1992). For this reason, we measured the activity of transcription factors: AP-1, CREB and NF-κB, by a reporter gene assay. Cyrneine A (200 μM) significantly increased the activity of AP-1 and NF-κB, but CREB activation was not observed (Fig. 5A). AP-1 activation is often accompanied by expression of immediate early genes, we therefore determined the relative mRNA levels of *c-fos* and *c-jun* by RT-PCR. Although the *c-jun* message level remained unaltered, a transient and dramatic increase in *c-fos* gene expression was observed in response to cyrneine A (200 μM) treatment, an effect comparable to that induced by PMA (Fig. 5B). Furthermore, CREB phosphorylation at Ser 133 is associated with enhanced activation of transcription (Lonze and Ginty, 2002), however, we did not detect enhanced phosphorylation of CREB at Ser133 in response to cyrneine A (200 μM) treatment (Fig. 5C).

In the process of neurite extension, rearrangement of cytoskeletal proteins such as actin is observed in addition to gene expression events (Yamaguchi et al., 2001). Therefore, we monitored cyrneine A-induced actin translocation (Fig. 6). After stimulation with cyrneine A for 10 min, cells extended short neurites and F-actin accumulated at the tip of neurites for at least 60 min (Fig. 6). Because Rac1 is responsible for neurite outgrowth induced by NGF (Aoki et al., 2004, 2005), we examined the involvement of Rac1 in cyrneine A-induced neurite outgrowth. When an interfering mutant of Rac1 (RacN17) and EGFP was overexpressed in PC12 cells, neurite outgrowth induced by cyrneine A (150 μM) was significantly blocked (Fig. 7A and B). EGFP alone did not affect cyrneine A-induced neurite outgrowth (data not shown). In addition, Rac1 activity was measured by a pull-down assay using GST-PAK-CRIB as a probe. NGF (100 ng/ml) led to a robust activation of Rac1 after 2 min, and Rac1 was moderately activated by cyrneine A (150 μM) at the two time points tested (5 and 30 min; Fig. 7C). Taken together, these results suggest that cyrneine A induces neurite outgrowth in a Rac1-dependent mechanism.

Discussion

In the present study, we have shown that cyrneine A induced neurite outgrowth in PC12 and NG108-15 cells. In order to understand how cyrneine A induces neurite outgrowth, we have investigated the role of signal transduction mechanisms that have previously been implicated in neurite extension. NGF induces a strong and sustained ERK activation that is required for neurite extension (Harada et al., 2001; Obara et al., 2004). Moreover, outgrowth has been shown to be dependent upon ERK pathway regulation of transcription (Markus et al., 2002). We investigated whether activation of the ERK pathway was responsible for the neurite outgrowth induced by cyrneine A. Cyrneine A did not induce activation of ERK (Fig. 3B and C), although PD98059 which is able to inhibit ERK 1/2 and 5, partially blocked the neurite extension induced by cyrneine A (Fig. 3A). Furthermore, the scabronines, which have a chemical structure similar to cyrneines, have been shown to activate the

atypical PKC isoform ζ in 1321N1 human astrocytoma cells (Obara et al., 2001). However, the involvement of PKC in cyrneine A-induced neurite outgrowth was excluded since the PKC inhibitor GF109203X did not block neurite extension (Fig. 3A) and PKC isoforms did not translocate from cytosolic to membrane fractions, an indicator of PKC activation, following treatment with cyrneine A (Fig. 4C). With regard to second messenger pathways, cAMP and IP_3 have been shown to promote neurite outgrowth in PC12 cells (Obara et al., 2002, 2004; Vossler et al., 1997), however, we did not observe significant accumulation of inositol phosphates nor cAMP (Fig. 4A and B) in response to cyrneine A treatment.

Rac1 and Cdc42 have been implicated in the formation of lamellipodia and filopodia, and they have been shown to positively regulate neurite outgrowth (Aoki et al., 2004, 2005). Conversely, RhoA inhibits NGF- or fibroblast growth factor-induced Rac1 activation or neurite outgrowth (Harada et al., 2005; Yamaguchi et al., 2001). Although ERKs or JNKs are downstream signaling effectors of Rac1 (Eblen et al., 2002; Lim et al., 1996), we did not observe activation of ERKs or JNKs in response to cyrneine A. However, we have shown that cyrneine A activates Rac1, as determined by pull-down assay (Fig. 7C). In addition, a dominant-negative Rac1 mutant inhibited cyrneine A-induced neurite extension (Fig. 7A and B). Therefore, Rac1 appears to be involved in cyrneine A-induced neurite outgrowth, as it is for NGF-induced neurite outgrowth. However, it remains to be established exactly how cyrneine A activates Rac1, and, as other G-proteins such as Rho A and Cdc42 regulate actin dynamics, it will be necessary to clarify their roles in this process too.

In PC12 cells, NGF activates Rac1 through Rac1 GEFs such as Vav 2/3 accompanied by the translocation to the cytoplasmic membranes. The local generation of phosphatidylinositol 3,4,5-trisphosphate, which is a major product of PI3K, is essential for recruiting Vavs to plasma membranes and, in addition, PI3K regulate ERK activation and neurite outgrowth in response to NGF (Aoki et al., 2004, 2005; York et al., 2000). As the PI3K inhibitor, wortmannin, did not inhibit neurite outgrowth induced by cyrneine A (Fig. 3A), we can reasonably assume that the PI3K-Vav pathway is not involved in the activation of Rac1 in response to cyrneine A. Src tyrosine kinase could be a potential candidate for signaling to Rac1 in the case of cyrneine A, since Src-mediated tyrosine phosphorylation of Vav 2 is essential for nectin-induced Rac1 activation and lamellipodia formation (Fukuyama et al., 2005).

The transcription factors AP-1 and NF-κB become activated following stimulation by a variety of growth factors, cytokines, neurotransmitters and during stress (O'Neill and Kaltschmidt, 1997; Shen et al., 2005). We have shown that cyrneine A leads to activation of AP-1 and NF-κB (Fig. 5), but the mechanism of activation remains to be established. Moreover, gene expression of proto-oncogene, *c-fos*, was robustly enhanced in response to cyrneine A.

To date, we have shown that both cyreines A and B have the ability to induce neurite outgrowth (Marcotullio et al., 2006), whereas cyrneines C and D do not (unpublished observation). Several cyathane diterpenes such as erinacines and scabronines have been shown to induce NGF in glial cells (Kawagishi et al., 1994; Obara et al., 1999, 2001). However, we have been unable to detect a direct effect of the scabronines at concentrations as high as 100 μM, on neurite outgrowth in PC12 cells: whether a higher concentration is required, remains to be determined, as do the effects of erinacines on neurite outgrowth. On the other hand, the effect of cyrneine A on NGF expression in glial cells is currently under investigation. It will be interesting to determine whether the variety of cellular responses to cyathanes (e.g. neurite outgrowth and NGF production) is attributed to the small differences in functional groups on cyathane structure. Clearly, it will be necessary to elucidate the structure–activity relationship of the cyathanes.

In conclusion, we have demonstrated that cyrneine A induces neurite outgrowth in PC12 cells via Rac1. Although the neurotrophic effect of cyrneine A in primary cultured neurons and in vivo is yet to be examined, the data presented here suggest that cyrneine A could be a promising lead compound in the development of therapeutic agents against neurodegenerative conditions such as Alzheimer's and Parkinson's disease.

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